

Network-Based Analysis

FIFA World Cup 2022

Teams as nodes. Results, performance and style as edges.

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Network Science Project

Research Focus

One tournament, multiple network interpretations

- Can network rankings reproduce competitive hierarchy?
- Which teams dominated statistically?
- Which teams were efficient, disruptive, or stylistically similar?

64

matches

32

teams

Analysis Pipeline

From match table to generated network outputs



Network Construction

Same teams, different edge meanings

- **Signed match** team1 -> team2, weight = goal difference
- **Outcome** loser -> winner; draws are reciprocal
- **Statistical dominance** stronger standardized match profile -> opponent
- **Offensive efficiency** better conversion ratios -> opponent
- **Pressure / recovery** more defensive pressure and forced turnovers -> opponent
- **Style similarity** teams linked by similar average profiles

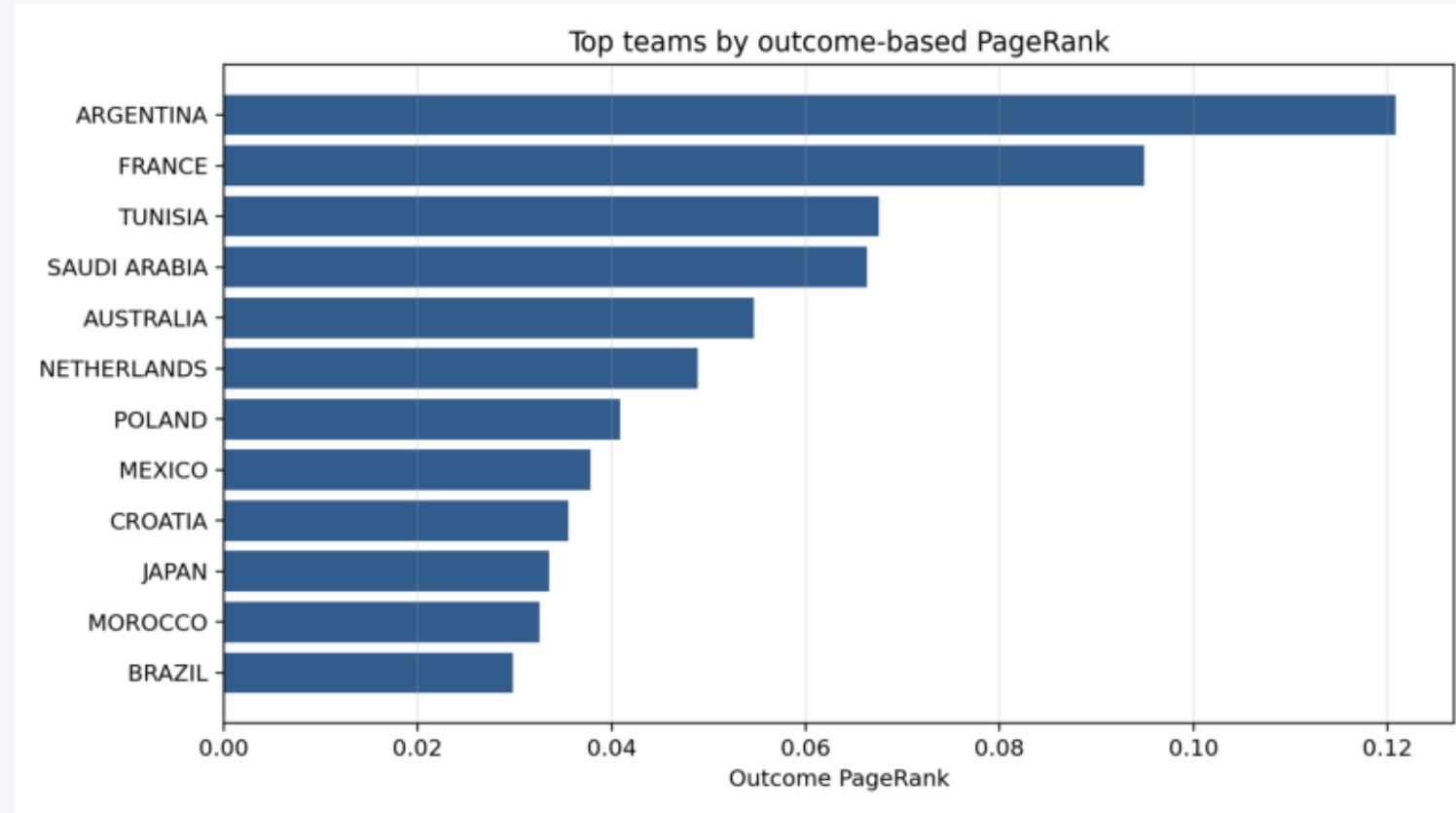
Tournament Context

Knockout bracket and final path



Outcome PageRank

Ranking success through opponent importance



Key points

- Argentina #1
- France #2
- Upsets gain structural value

PageRank Upset Context

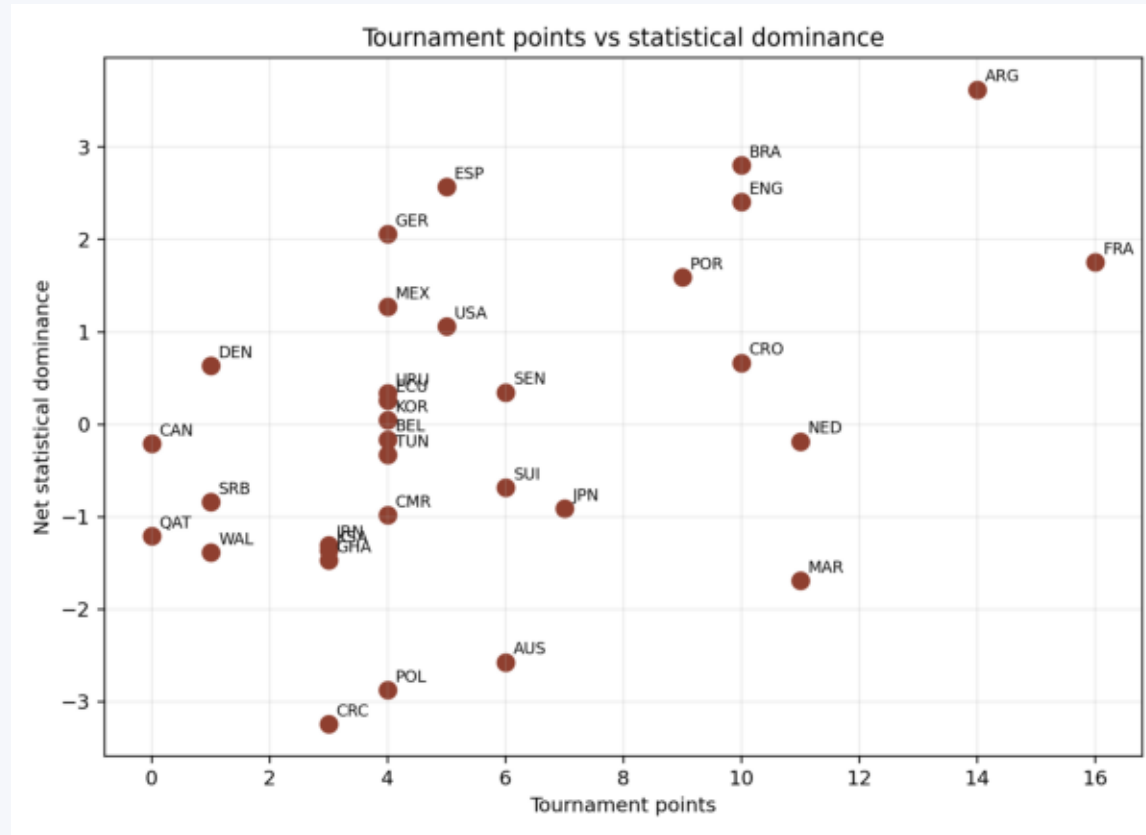
Why Tunisia and Saudi Arabia appear unusually high



- Saudi Arabia receives result support from Argentina.
- Tunisia receives result support from France.
- Because Argentina and France are important nodes, these upset links carry extra PageRank value.

Statistical Dominance

Match control is not the same as the points table

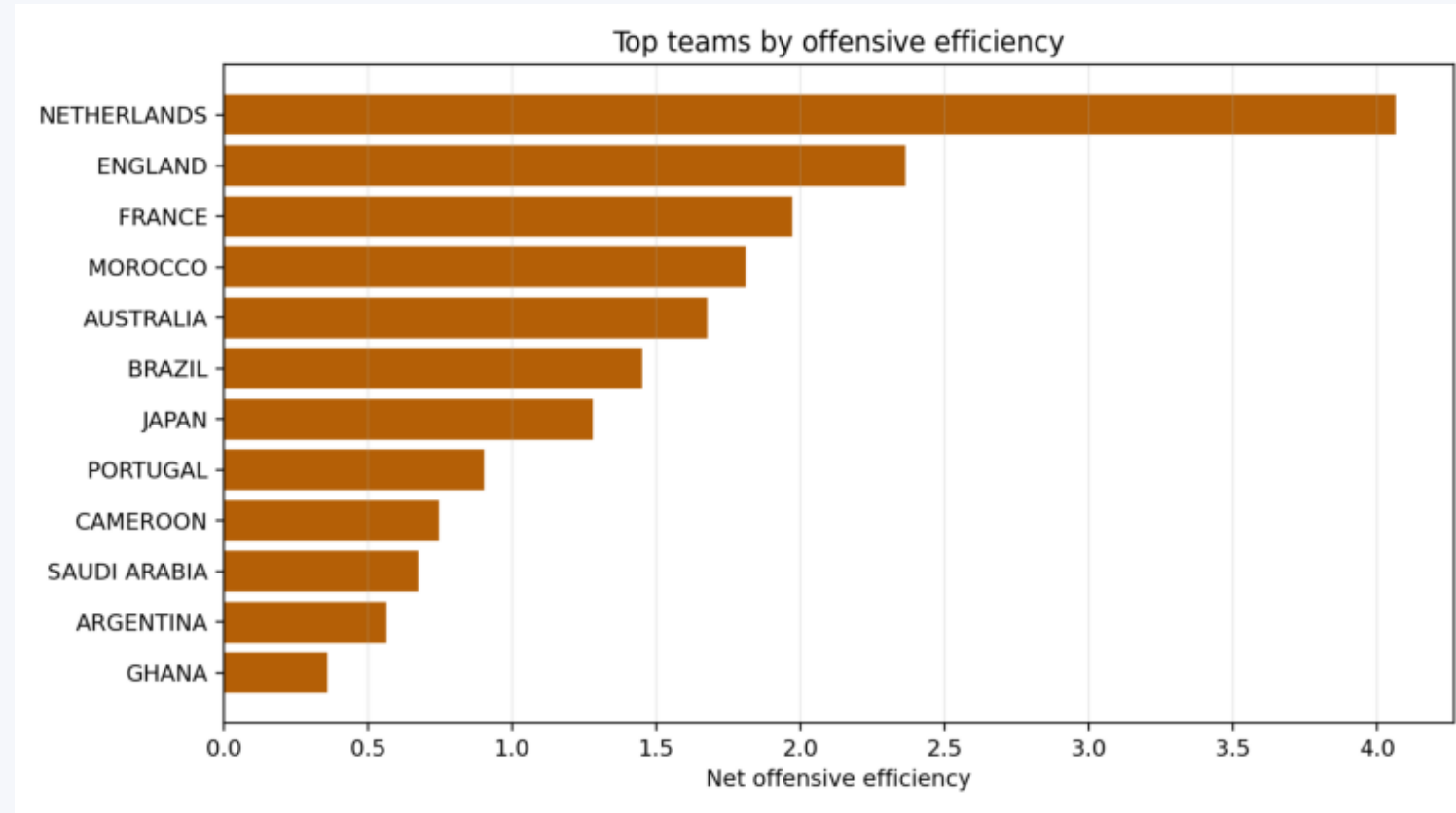


Key points

- Argentina leads
- Brazil, Spain, England high
- Dominance tracks control

Offensive Efficiency

Conversion over volume



Key points

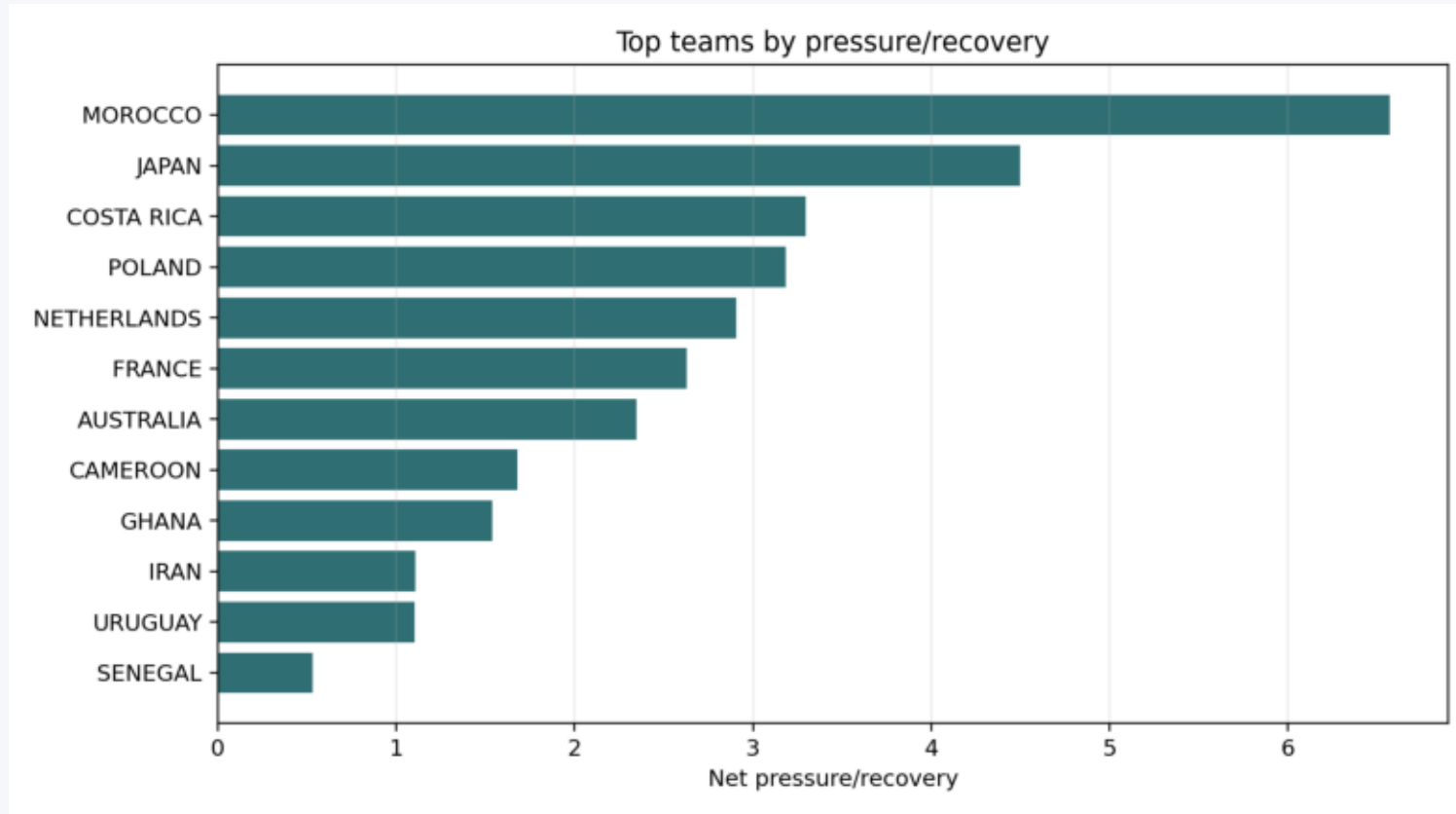
- Netherlands #1
- England and France high
- Rewards finishing efficiency

Pressure / Recovery

Defensive disruption and regain actions

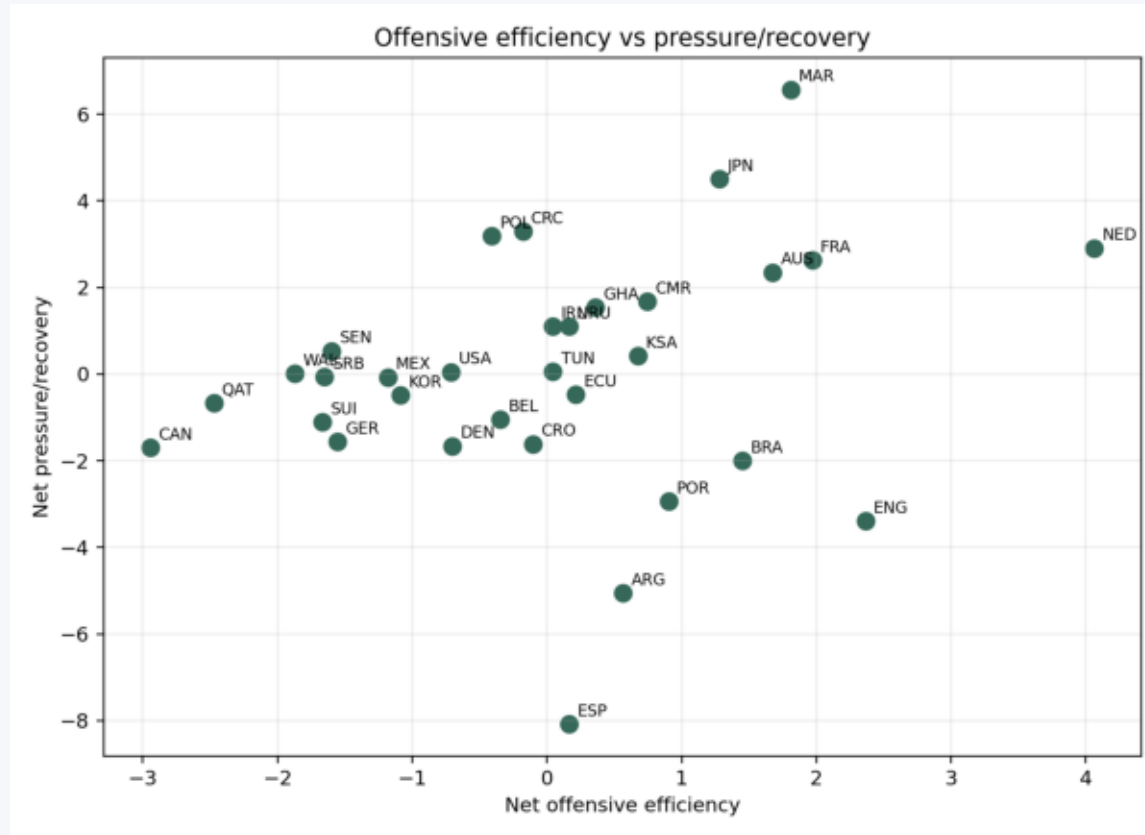
Key points

- Morocco #1
- Japan #2
- Different from possession dominance



Efficiency vs Pressure

Two distinct performance profiles

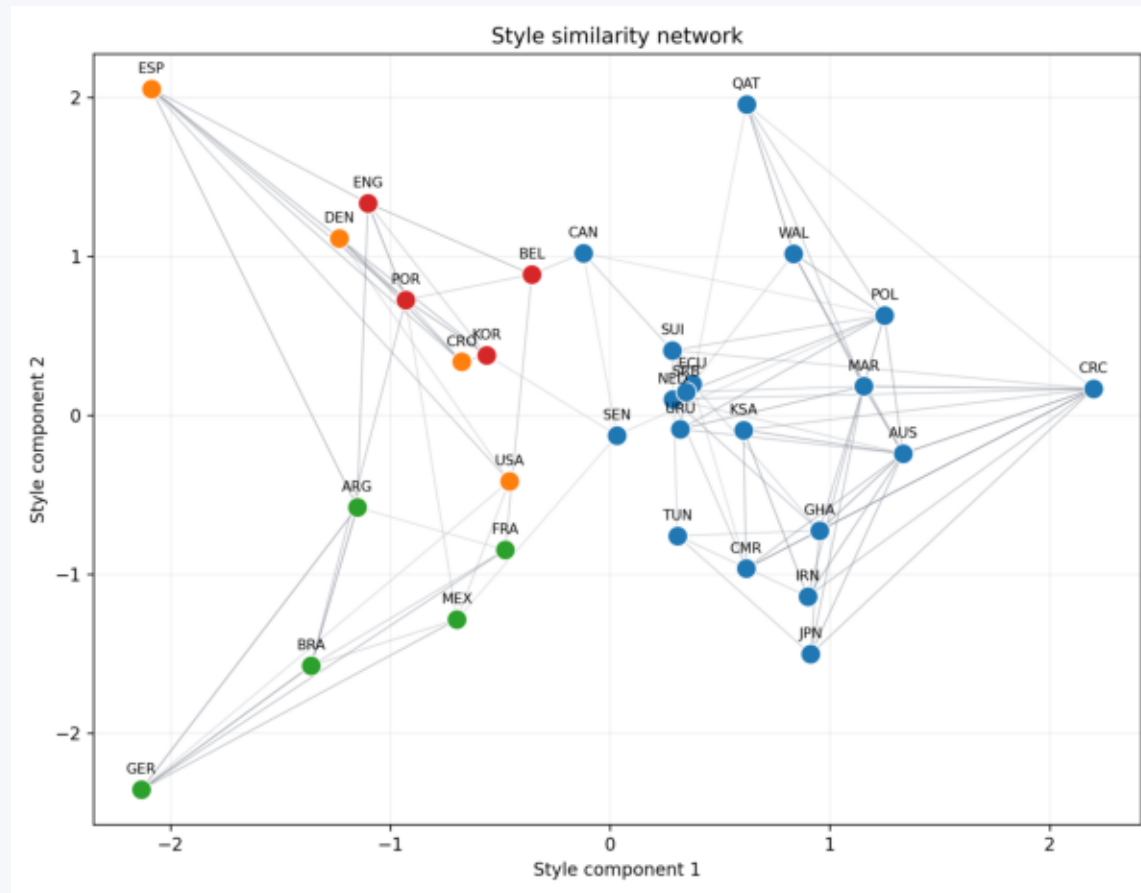


Key points

- Efficiency aligns with goals
- Pressure identifies disruption
- Morocco/Japan stand out

Style Similarity Network

Teams linked by similar average match profiles

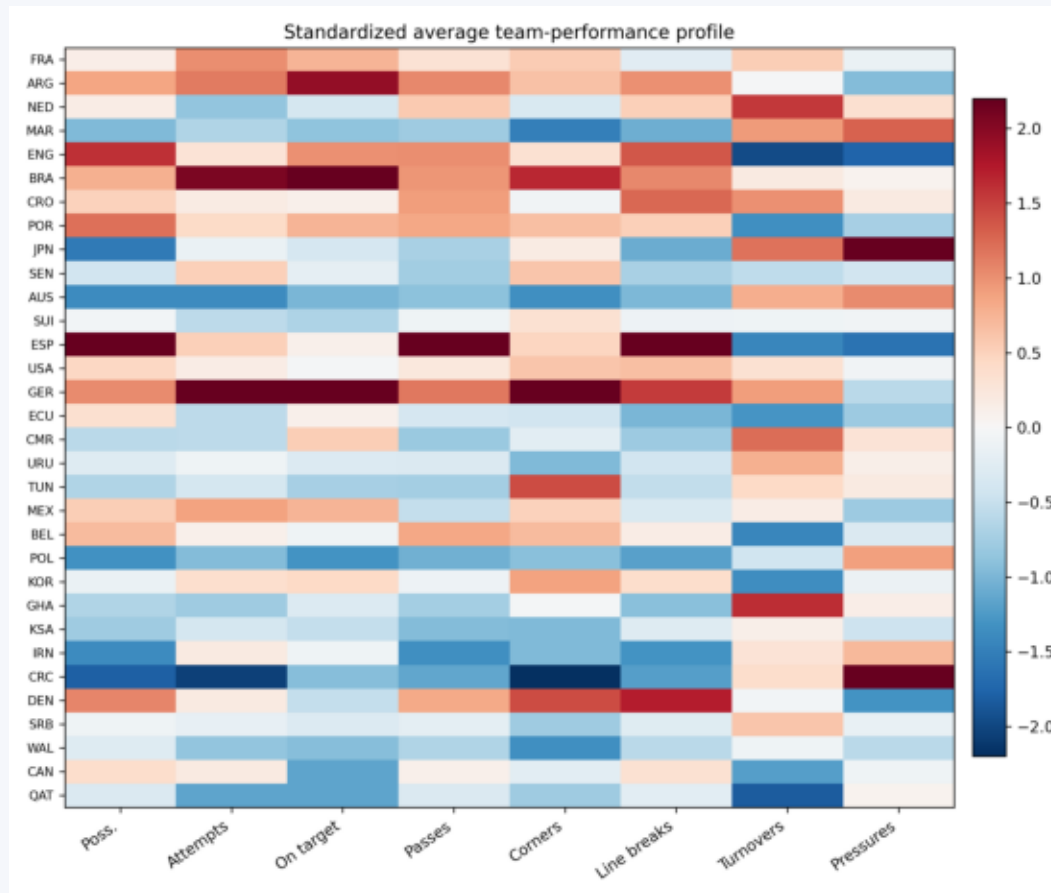


Key points

- Cosine similarity
- 4 nearest neighbours
- Label propagation communities

Performance Heatmap

Standardized team profiles



Key points

- Spain: possession/passes
- Brazil/Germany: shot volume
- Morocco/Japan: pressure

Main Takeaway

Changing the edge definition changes the football story

Results, dominance, efficiency, pressure and style are related, but not equivalent.

- Outcome PageRank captures opponent-sensitive success.
- Performance networks separate control, conversion and disruption.
- Style similarity groups teams by how they played.